

Bayesian Bivariate Markov Switchg Multifractal Model

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1 Introduction

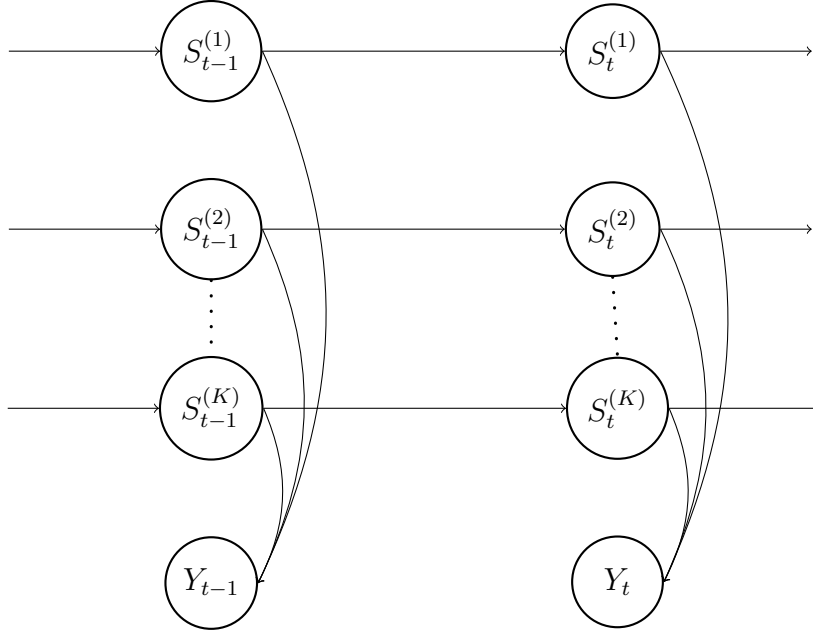
Markov Switching Multifractal model (MSM) is a stochastic volatility model built exclusively for financial application, and as far as I know of is only utilized in this field. However, its characteristics can be generalized statistically, and is therefore open to expansion which is exactly what this project aims to bring.

2 Original Model

2.1 Factorial Hidden Markov Models

Statistically, the univariate MSM first introduced in Calvet and Fisher (2001) [1] is a specific implementation of Factorial Hidden Markov Models (FHMM) as conceptualized by Ghahramani and Jordan (1997) [3]. The FHMM has a following Directed Acyclic Graph (DAG) representation:

Figure 1: DAG of FHMM with K states at any given time



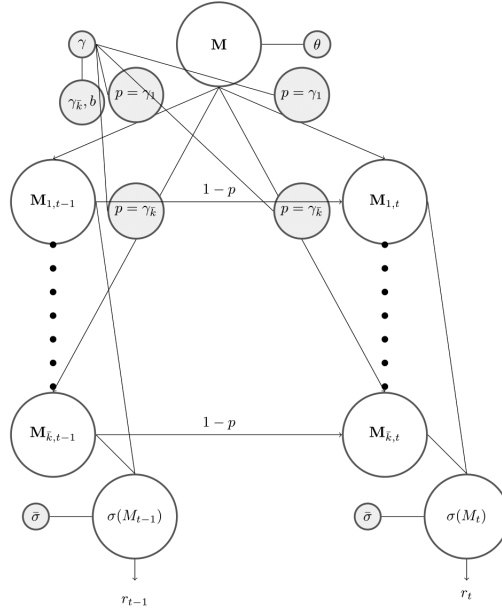
Where states $S_{1:T}^{(i)}$ at any given layer $i = 1, 2, \dots, K$ have Markov property with respect to other states within its layer. Then, for each time t , these states will jointly yield observations Y_t . This general formulation of FHMM has nice property such as the joint distribution of these states at any given time t also possesses Markov property as well as availability of exact likelihood.

Fundamentally, the univariate MSM belongs to this family of models with specific state transitions and joint generative structure of observations which I will briefly introduce in next subsection.

2.2 Univariate Markov Switching Multifractal Model

The univariate MSM has a following DAG representation:

Figure 2: DAG of MSM



Even without the knowledge of these parameters, we can see the similarity of it to FHMM. The description of these parameters are as follows:

- $\bar{k}$: Number of state layers at any given time which corresponds to K in the FHMM.
- $M_{i,t}$: State parameters with Markov property which corresponds to S_t^i in the FHMM.
- r_t : Observed log returns used as response variable which corresponds to Y_t in the FHMM.
- γ_1 : Switching probability which coupled with b controls the dynamics of $M_{i,t}$ s by defining $\gamma_1, \dots, \gamma_{\bar{k}}$.
- b : Spacing parameter which controls the difference in magnitude between γ_i and γ_{i+1} .
- $\bar{\sigma}$: Scaling parameter of the volatility. $\sigma(M_t)$ is a function of $\bar{\sigma}$ and $M_{i,t}$ s which yields r_t .
- M, θ : The distribution and corresponding list of parameters where state realizations (the value each $M_{i,t} \forall i, t$ can take) are drawn.

In total, there are 4 parameters to be estimated (or sampled from posterior since we do it in Bayesian way) which we define $\phi \in \{\theta, \gamma_1, b, \bar{\sigma}\}$. In addition, $\bar{k}$ is a model parameter set by the user where making $\bar{k}$ greater will allow the model to have greater variation in r_t s which depending on the observed variations of r_t s could be either good or bad.

Then, the probabilistic dynamics of these parameters are defined as follows:

$$\begin{aligned}
 r_t &\sim N(0, \sigma(M_t)) \text{ where } \sigma(M_t) \equiv \bar{\sigma}^2 \prod_{i=1}^{\bar{k}} M_{i,t} \\
 M_{i,t} &= M_{i,t-1} + \mathbb{1}_{u < \gamma_i} (\bar{m} - M_{i,t-1}), & u &\sim U(0, 1) \\
 \bar{m} &\sim M(\theta) \text{ where } M(\theta) \equiv m_0 \mathbb{1}_{u' < 0.5} + (2 - m_0) \mathbb{1}_{u' \geq 0.5}, & u' &\sim U(0, 1) \\
 \text{thus } \theta &\equiv \{m_0\} \text{ for our case called binomial MSM} \\
 \gamma_i &= 1 - (1 - \gamma_1)^{(b^i - 1)}, & i &= 1, \dots, \bar{k}
 \end{aligned}$$

Which fully describes the overall generative structure of the univariate MSM. The original bivariate MSM which just adds a simple tweak to this univariate specification to model two series jointly.

2.3 Bivariate Markov Switching Model

Calvet et al (2006) [2] introduced the Bivariate version of this model by introducing dependence in switches and observation equation. That is, if we have two series α, β , following the univariate specification we now have states $M_{i,t}^\alpha, M_{i,t}^\beta$ which experience switches through variables $1_{i,t}^\alpha \in \{0, 1\}, 1_{i,t}^\beta \in \{0, 1\}$ similar to $1_{u < \gamma_i} \in \{0, 1\}$ in univariate case above.

Then they introduced simple conditional dynamics of switches as follows:

$$p(1_{i,t}^\beta = 1) = \gamma_i$$

$$p(1_{i,t}^\alpha = 1 | 1_{i,t}^\beta = 1, \lambda) = (1_{i,t}^\beta - \lambda)\gamma_k + \lambda \text{ where } \lambda \in [0, 1]$$

where $\lambda = 1$ cases corresponds to complete independence of switches of α to that of β at layer i and $\lambda = 0$ to complete dependence. The dependence in observation equation is given as follows:

$$\begin{pmatrix} r_t^\alpha \\ r_t^\beta \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma(M_t^\alpha) & \rho \sqrt{\sigma(M_t^\alpha)\sigma(M_t^\beta)} \\ \rho \sqrt{\sigma(M_t^\alpha)\sigma(M_t^\beta)} & \sigma(M_t^\beta) \end{pmatrix} \right] \text{ where } \sigma(M_t^\alpha) = \bar{\sigma}^\alpha \prod_{i=1}^{\bar{k}} M_{i,t}^\alpha$$

and ρ representing correlation of $\sigma(M_t)$ s.

2.4 Issues with Their original Bivariate Specification

Although their specification to extend univariate MSM to Bivariate model is succinct, assuming that switches in α, β to be dependent only through same layers is quite restrictive. Also due to this constraint, $\bar{k}$ has to be set to same numbers for both α, β which does not make sense if we are modeling two series with entirely different variations but yet may have some structural similarity we want to capture by modeling jointly.

For example, commodity spot price and futures series are by construction heavily dependent with each other. However, due to futures being more sensitive to speculations, the volatility of futures are generally higher than that of spot prices. Therefore, when we are modeling these series jointly, we would like to have granularity of our model to be higher on futures series than spot prices series by assigning different $\bar{k}$ s to these series and capture dependence of switches across layers between these two.

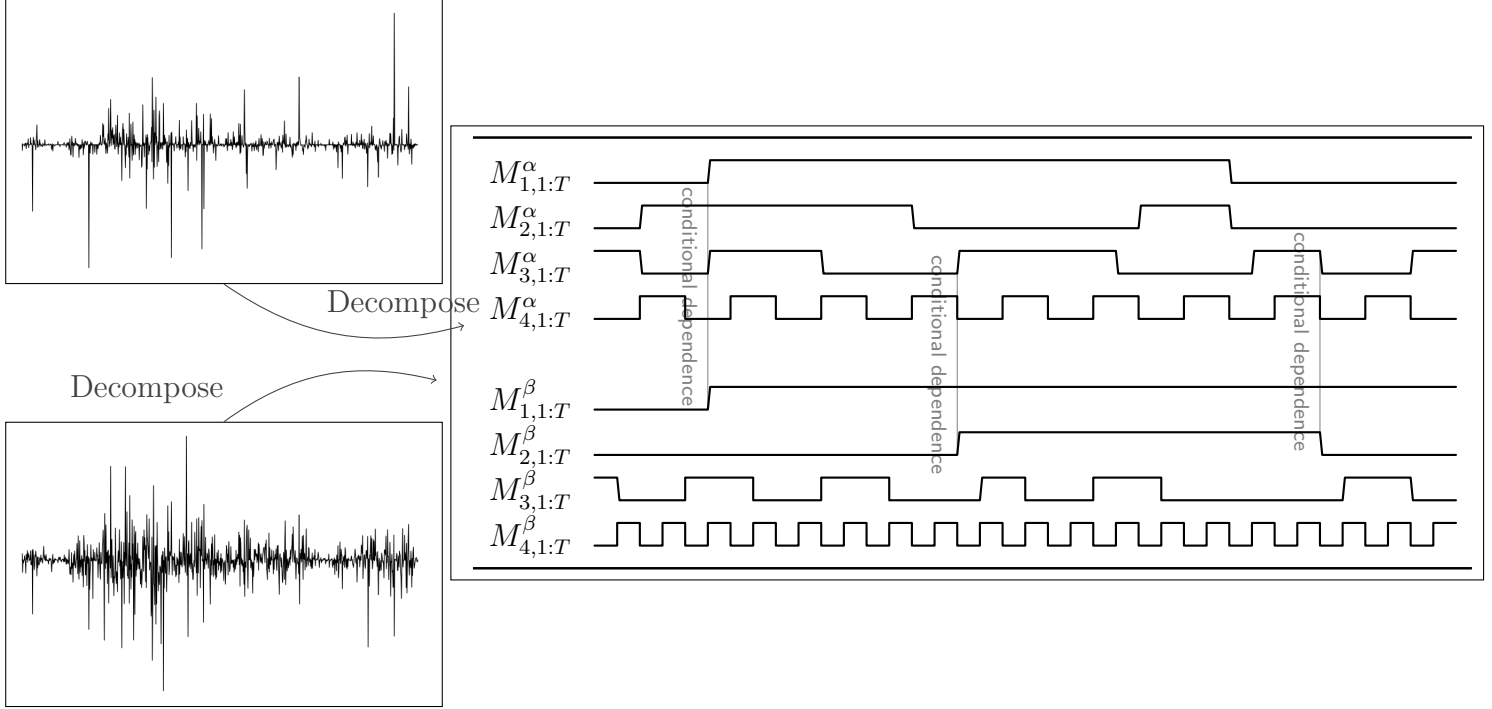
The model we developed called Bayesian Bivariate Markov Switching Multifractal model addresses this very issue and rephrases the entire model in Bayesian specification.

3 Bayesian Bivariate Markov Switching Multifractal Model

3.1 Intuition of the Model

Essentially, we want our model to decompose each mean-zero volatility series into several layers of Markov Switching Processes $M_{i,1:T}^\alpha$ and $M_{i',1:T}^\beta$ (does not have to be same number of layers as in the case of original Bivariate MSMS) and capture conditional dependence of switches in layers across series as depicted in the next page.

Figure 3: Model Intuition



As we can see, a switch in first layer of α series may have conditional dependence to that of first layer of β which can also be captured by the original model. However notice that switches in 3rd layer of α series may also have dependence to the same switch in β which the original model will not capture due to the aforementioned restriction. Additionally, we also observe strong conditional dependence of switches in 3rd layer of α series that that of the 2nd layer in β .

If we expect this example to be realistic (which we do), an **interfrequency approach** to not necessarily limit only the switches in the same frequency to be dependent across series is needed. Our model provides this exact feature by taking advantage over more flexible modeling constraints in Bayesian statistics.

3.2 Model Specification

For our bivariate case where we have two series α, β , there are pairs of univariate parameters $\{(m_0^\alpha, m_0^\beta), (\gamma_1^\alpha, \gamma_1^\beta), (b^\alpha, b^\beta), (\bar{\sigma}_\alpha, \bar{\sigma}_\beta)\}, (\bar{k}^\alpha, \bar{k}^\beta)$ as well as correlation ρ and latent variables called switches $1_{1:\bar{k}^\alpha, t}^\alpha \in \{0, 1\}^{\bar{k}^\alpha}, 1_{1:\bar{k}^\beta, t}^\beta \in \{0, 1\}^{\bar{k}^\beta}$ and coefficient for these switches $B_{1:\bar{k}^\alpha, 1:\bar{k}^\beta} \in \{0, 1\}^{(\bar{k}^\alpha \times \bar{k}^\beta)}$. Then, by specifying conditional dynamics of α given β by regression, we have a joint model for bivariate returns as follows:

$$\begin{pmatrix} r_t^\alpha \\ r_t^\beta \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha & \rho \sqrt{\bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha} \sqrt{\bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta} \\ \rho \sqrt{\bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha} \sqrt{\bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta} & \bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta \end{pmatrix} \right]$$

$$M_{i,t}^\beta = M_{i,t-1}^\beta + 1_{i,t}^\beta (\bar{m}^\beta - M_{i,t-1}^\beta), \quad i = 1, \dots, \bar{k}^\beta, \quad 1_{i,t}^\beta \sim \text{Bernoulli}(\gamma_i^\beta)$$

$$M_{i,t}^\alpha = M_{i,t-1}^\alpha + 1_{i,t}^\alpha (\bar{m}^\alpha - M_{i,t-1}^\alpha), \quad i = 1, \dots, \bar{k}^\alpha, \quad 1_{i,t}^\alpha \sim \text{Bernoulli}(\pi_i)$$

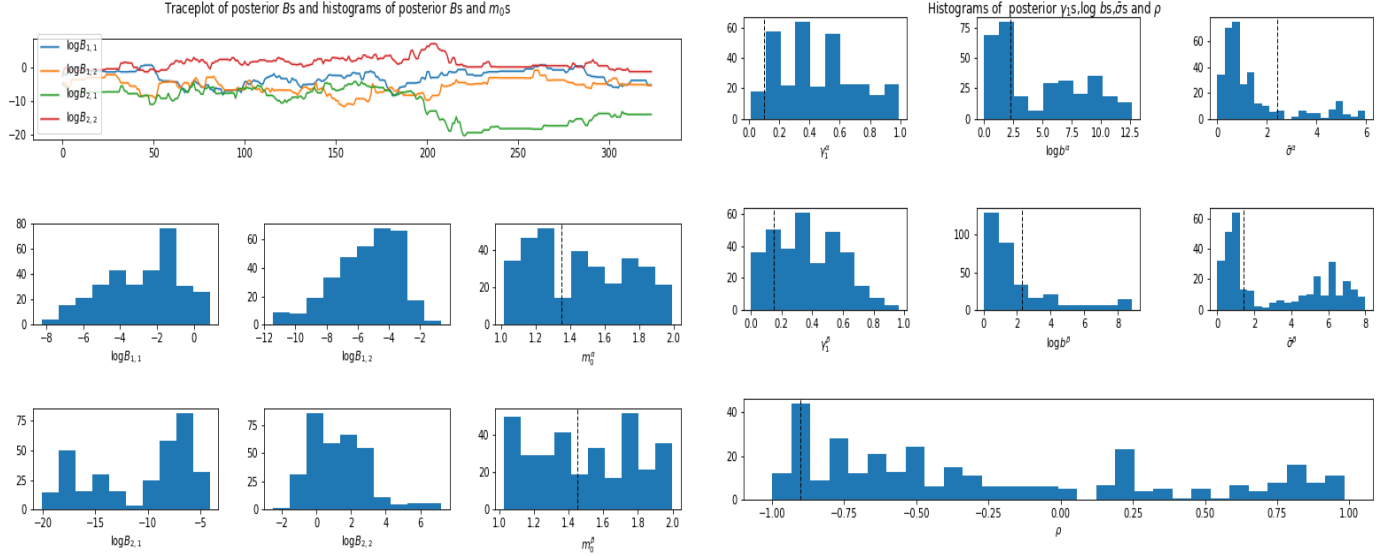
$$\text{logit}(\pi_i) = C + B_{i,1} 1_{1,t}^\beta + \dots + B_{i,\bar{k}^\beta} 1_{\bar{k}^\beta, t}^\beta, \quad i = 1, \dots, \bar{k}^\alpha, \quad C = \text{logit}(\gamma_i^\alpha) - B_{i,1} \gamma_1^\beta - \dots - B_{i,\bar{k}^\beta} \gamma_{\bar{k}^\beta}^\beta$$

where we can trivially check that π_i when marginalizing out $1_{1:\bar{k}^\beta, t}^\beta$ is γ_i^α . By setting appropriate priors to B s we can recover sparse structure of B s where elements with considerable nonzero posterior samples

corresponds to conditional dependence between that particular layer in β and α . The likelihood of this model is analytically computable by distributed computing (not shown why for space constraints) and we can run MCMC on these parameters (not shown as well) due to this availability of the likelihood.

3.3 Results on Synthetic Data

To verify the feasibility of our sampling algorithm, we generated synthetic observations following this model and here are our results.



Here, we simulated data that follows MSM with $\bar{k} = (2, 2)$ and only $B_{2,1}$ being nonzero, meaning 2nd layer Markov Switching Processes of α is conditionally dependent on switches in 1st layer of β .

Resulting posterior draws from our MCMC suggest some mixing issues. Still, posterior distributions of some parameters are already concentrated around the true value (horizontal dotted line for each histogram).

The interesting part is that although the result of the traceplot of B_s do suggest that there are only one non-zero components (remember that it's in log scale), it somehow recovered $B_{2,2}$ to be that component rather than $B_{2,1}$ in the true model. Thus it did manage to find that second layer of α is dependent on switches in one layer of β but missed the specific layer.

3.4 Appendix

This project funded by BEST foundation was done with teammate Jaeyeon Lee, Mentored by Prof. Ma and West. To find all of our implemented algorithms jump to https://github.com/Jantg/BEST_proj

References

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